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EED

All Circuit Theorems

- Superposition Theorem
- Thevenin's Theorem
- Norton's Theorem
- Maximum Power Transfer
- Exercise

Superposition Theorem

- If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to use nodal or mesh analysis.
- Another way is to determine the contribution of each independent source to the variable and then add them up. The latter approach is known as the *superposition*.
- The idea of superposition rests on the linearity property.

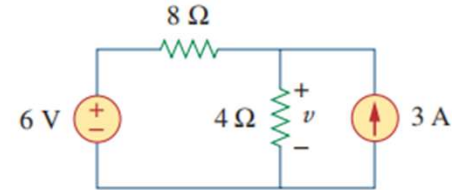
Statement: The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

Steps to Apply Superposition Principle:

1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using the techniques.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

Example:

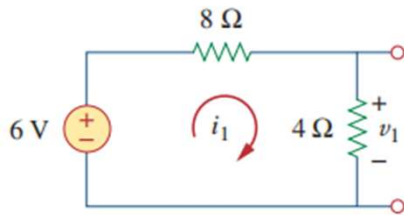
Use the superposition theorem to find v in the circuit of Fig.



Solution: Since there are two sources, let

$$v = v_1 + v_2$$

where v_1 and v_2 are the contributions due to the 6-V voltage source and the 3-A current source, respectively. To obtain v_1 we set the current source to zero, as shown in Fig. (a). Applying KVL to the loop in



(a)

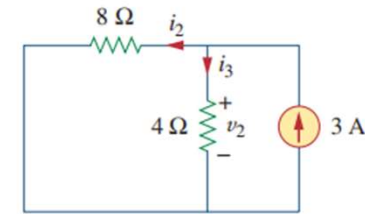
$$12i_1 - 6 = 0 \quad \Rightarrow \quad i_1 = 0.5 \text{ A}$$

Thus,

$$v_1 = 4i_1 = 2 \text{ V}$$

We may also use voltage division to get v_1 by writing

$$v_1 = \frac{4}{4 + 8}(6) = 2 \text{ V}$$



(b)

To get v_2 , we set the voltage source to zero, as in Fig. (b). Using current division,

$$i_3 = \frac{8}{4 + 8}(3) = 2 \text{ A}$$

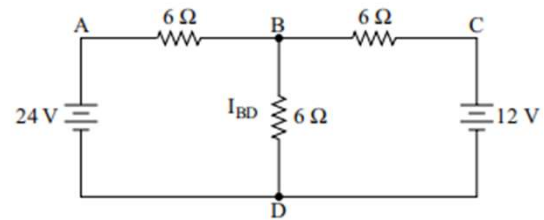
Hence,

$$v_2 = 4i_3 = 8 \text{ V}$$

And we find

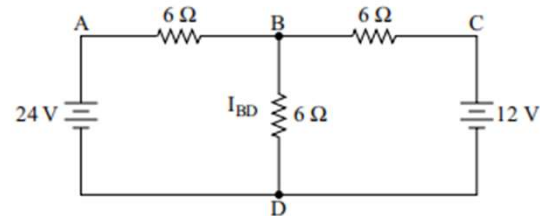
$$v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

Example: Using the superposition theorem find the value of current, I_{BD} in the circuit shown in Fig.



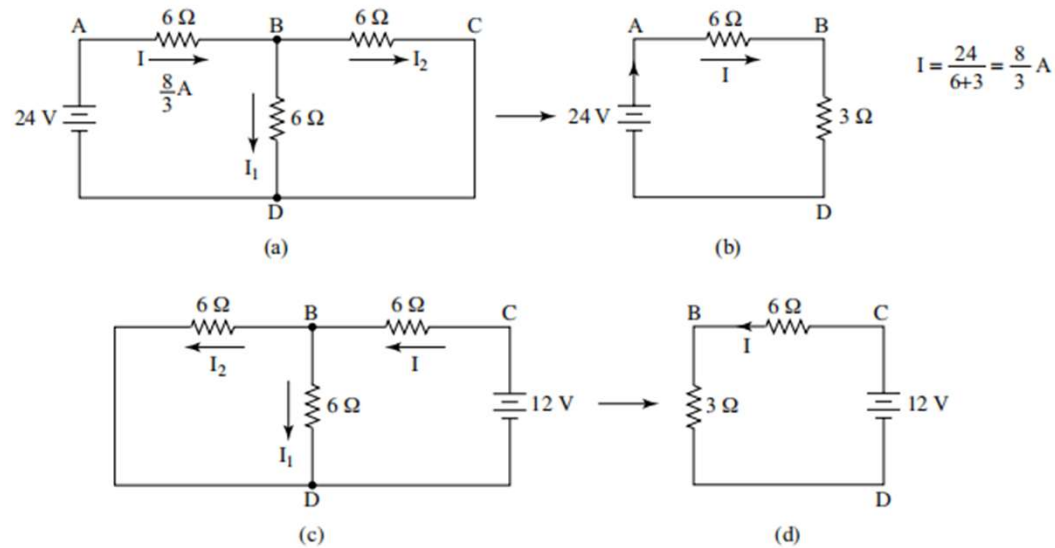
Solution:

Example: Using the superposition theorem find the value of current, I_{BD} in the circuit shown in Fig.



Solution:

We shall consider each source separately and calculate the current flowing through the branch BD. First the 24 V source is taken by short circuiting the 12 V source as shown in Fig



The current flowing from the battery is calculated $I = \frac{8}{3} \text{ A}$ as shown in Fig. (b). This $I = \frac{8}{3} \text{ A}$ gets divided into two parts as I_1 and I_2 . Current through the resistor across BD is I_1 . To find I_1 we can use the current division rule as

$$I_1 = I \frac{R_2}{R_1 + R_2} = \frac{8}{3} \times \frac{6}{6+6} = \frac{4}{3} \text{ A}$$

Now, consider the 12 V source and short circuit the 24 V source as shown in Fig. (c). The current supplied by the 12 V source is calculated as

$$\begin{aligned} I &= \frac{12}{6+3} \\ &= \frac{4}{3} \text{ A} \end{aligned}$$

The total current I due to 12 V supply has been calculated as $\frac{4}{3} \text{ A}$. This current gets divided into I_1 and I_2 as has been shown in Fig. (a). Current I_1 is calculated using the current division rule as

$$I_1 = I \times \frac{R_2}{R_1 + R_2} = \frac{4}{3} \times \frac{6}{6+6} = \frac{2}{3} \text{ A}$$

To determine the current flowing through the resistor across BD, the combined effect of the two voltage sources will be taken. Therefore,

$$\begin{aligned} I_{BD} &= I_1 \text{ due to 24 V source} + I_1 \text{ due to the 12 V source} \\ &= \frac{4}{3} + \frac{2}{3} = 2 \text{ A, flowing from node B towards node D} \end{aligned}$$

Thevenin's Theorem

- It often occurs in practice that a particular element in a circuit is variable (usually called the *load*) while other elements are fixed. As a typical example, a household outlet terminal may be connected to different appliances constituting a variable load. Each time the variable element is changed, the entire circuit has to be analyzed all over again.
- To avoid this problem, Thevenin's theorem provides a technique by which the fixed part of the circuit is replaced by an equivalent circuit.

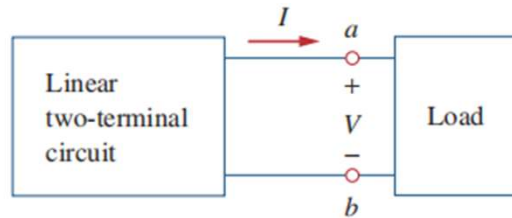


Fig: 1 (a) Original Circuit

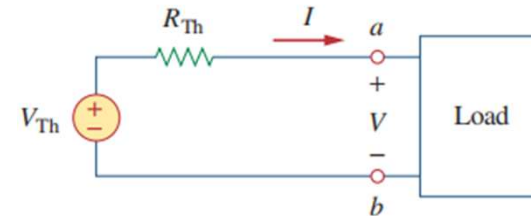


Fig: 1 (b) The Thevenin Eq. Circuit

- According to Thevenin's theorem, the linear circuit in Fig.(a) can be replaced by that in Fig.(b). The circuit to the left of the terminals in Fig.(b) is known as the *Thevenin equivalent circuit*; it was developed in 1883 by M. Leon Thevenin (1857–1926), a French telegraph engineer.

Statement: Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

The proof of the theorem our major concern right now is how to find the Thevenin equivalent voltage and resistance . To do so, suppose the two circuits in Fig. 1 are equivalent. Two circuits are said to be *equivalent* if they have the same voltage-current relation at their terminals. Let us find out what will make the two circuits in Fig. 1 equivalent. If the terminals are made open-circuited (by removing the load), no current flows, so that the open-circuit voltage across the terminals in Fig. (a) must be equal to the voltage source in Fig.(b), since the two circuits are equivalent. Thus is the open-circuit voltage across the terminals as shown in Fig. 2; that is,

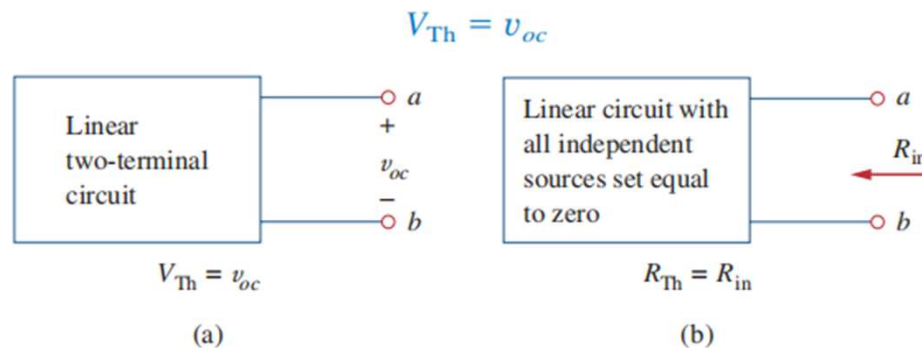


Fig: 2(a), (b) Finding the V_{th} and R_{th}

Again, with the load disconnected and terminals open circuited, we turn off all independent sources. The input resistance (or equivalent resistance) of the dead circuit at the terminals in Fig. 1(a) must be equal to in Fig. 1 (b) because the two circuits are equivalent. Thus, is the input resistance at the terminals when the independent sources are turned off, as shown in Fig. 2(b); that is,

$$R_{Th} = R_{in}$$

- Thevenin's theorem is very important in circuit analysis. It helps simplify a circuit. A large circuit may be replaced by a single independent voltage source and a single resistor. This replacement technique is a powerful tool in circuit design.
- As mentioned earlier, a linear circuit with a variable load can be replaced by the Thevenin equivalent, exclusive of the load. The equivalent network behaves the same way externally as the original circuit.
- Consider a linear circuit terminated by a load, as shown in Fig. 3(a). The current through the load and the voltage across the load are easily determined once the Thevenin equivalent of the circuit at the load's terminals is obtained, as shown in Fig. 3(b)

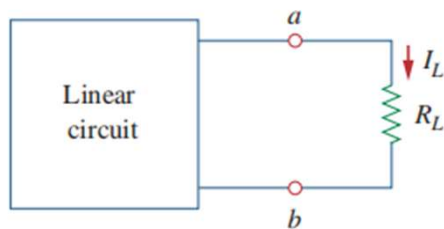


Fig:3(a) Original Circuit

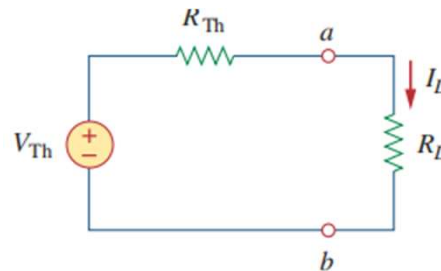
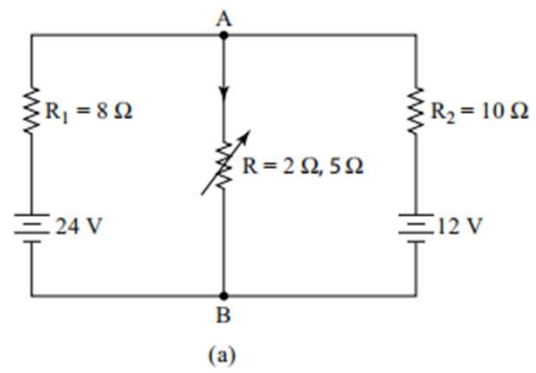


Fig:3(b) Thevenin Equivalent

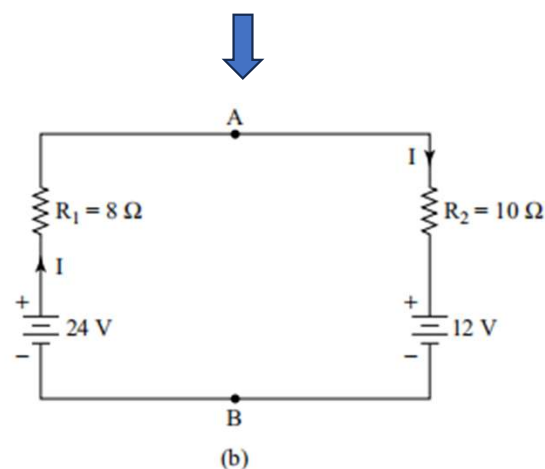
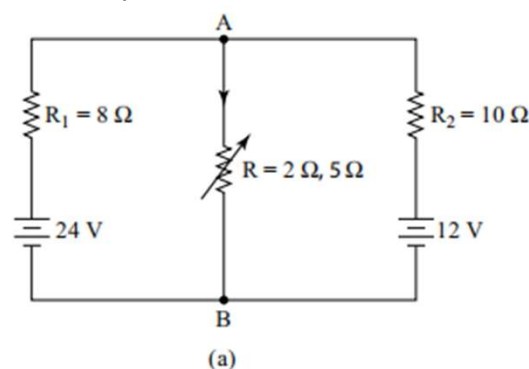
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

Example:



Example:



To calculate the current through the variable resistor R , the first step would be to take away the resistor R and calculate the V_{oc} across terminals A and B . Applying KVL in the circuit of Fig. (b)

$$+24 - 8I - 10I - 12 = 0$$

or,

$$I = 0.67 \text{ A}$$

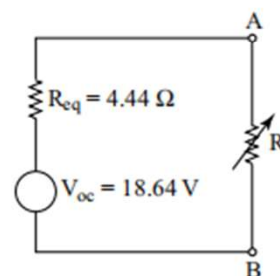
Approaching from B to A

$$V_{oc} = 24 - 8 \times 0.67 = 18.64 \text{ V}$$

The same result will be arrived at when we approach from A to B .

Now we calculate R_{eq} across terminals A and B by short circuiting the voltage sources or by replacing them by their internal resistances. Across terminals AB we find two resistances of 8Ω and 10Ω connected in parallel. R_{eq} across terminals A and $B = \frac{8 \times 10}{8 + 10} = 4.44 \Omega$

Thus, Thevenin's equivalent circuit is represented as in Fig. 2.50



Current through R with its variable values i.e., for $R = 2 \Omega$ and $R = 5 \Omega$ are calculated as

$$I = \frac{V_{oc}}{R_{eq} + R} = \frac{18.64}{4.44 + 2} = 2.89 \text{ A}$$

and

$$I = \frac{V_{oc}}{R_{eq} + R} = \frac{18.64}{4.44 + 5} = 1.97 \text{ A}$$

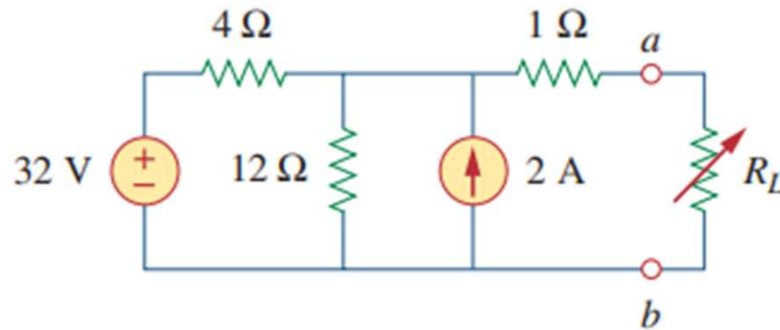
The steps involved in applying Thevenin's theorem in calculating current in a circuit component are as follows:

1. Remove the resistance from the circuit terminals through which current is to be determined.
2. Determine the open-circuit voltage that would be appearing across the circuit terminals wherefrom the resistance has been removed. This is called V_{oc} .
3. Calculate the equivalent resistance of the whole circuit across the terminals after replacing the sources of EMFs by their internal resistances (or by simply short-circuiting them if internal resistance is not provided or not known) and by keeping the current sources open-circuited (i.e., considering having infinite resistance).
4. Draw the Thevenin's equivalent circuit with V_{oc} as the voltage source, R_{eq} as the internal resistance of the voltage source, and R is the load resistance connected across the voltage source.
5. Calculate the current through R using the relation.

$$I = \frac{V_{oc}}{R_{eq} + R}$$

Exercise :

Find the Thevenin equivalent circuit of the circuit shown in Fig., to the left of the terminals. Then find the current through $R_L = 6, 16,$ and 36 ohm

Solution:

Norton's Theorem

In 1926, about 43 years after Thevenin published his theorem, E. L. Norton, an American Engineer at Bell Telephone Laboratories, proposed a similar theorem.

Statement: Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

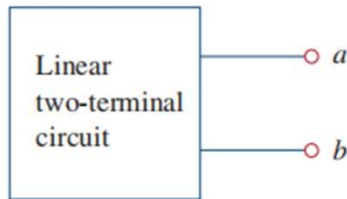
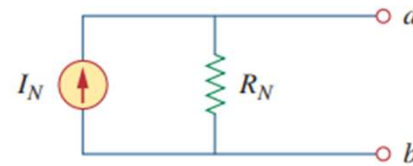


Fig: (a) Original Circuit Fig:



(b) Norton's Equivalent Circuit

The proof of Norton's theorem will be given in the next section. For now, we are mainly concerned with how to get R_N and I_N . We find R_N in the same way we find R_{th} . In fact, from what we know about source transformation, the Thevenin and Norton resistances are equal; that is,

$$R_N = R_{Th}$$

.....(1)

To find the Norton current we determine the short-circuit current flowing from terminal a to b in both circuits in Fig. It is evident *That* the short-circuit current in Fig.(b) is This must be the same short-circuit current from terminal a to b in Fig. (a), since the two circuits are equivalent. Thus,

$$I_N = i_{sc}$$

Observe the close relationship between Norton's and Thevenin's theorems: as in Eq. (1), and

$$I_N = \frac{V_{Th}}{R_{Th}}$$

.....(2)

This is essentially source transformation. For this reason, source transformation is often called Thevenin-Norton transformation.

Since V_{th} , I_N and R_{th} are related according to Eq. 2, to determine the Thevenin or Norton equivalent circuit requires that we find:

- The open-circuit voltage across terminals a and b .
- The short-circuit current at terminals a and b .
- The equivalent or input resistance at terminals a and b when

all independent sources are turned off. We can calculate any two of the three using the method that takes the least effort and use them to get the third using Ohm's law

$$V_{Th} = v_{oc}$$

$$I_N = i_{sc}$$

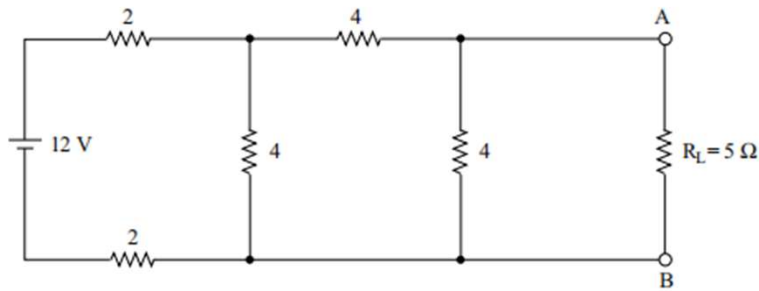
$$R_{Th} = \frac{v_{oc}}{i_{sc}} = R_N$$

What is the difference between Thevenin and Norton theorems?

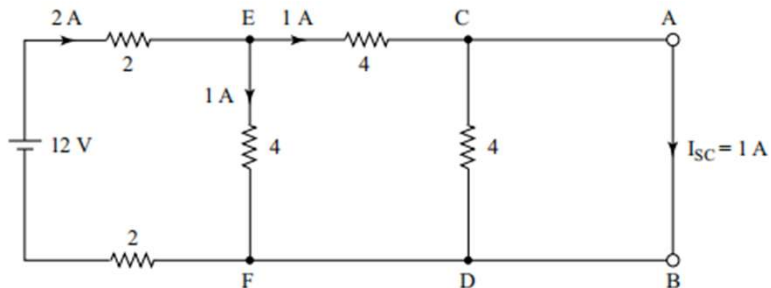
- – Norton's theorem uses a current source, whereas Thevenin's theorem uses a voltage source.
- – Thevenin's theorem uses a resistor in series, while Norton's theorem uses a resistor set in parallel with the source.
- – Norton's theorem is actually a derivation of the Thevenin's theorem.
- – The Norton's resistance and Thevenin's resistance are equal in magnitude.
- – The Norton's equivalent circuit and Thevenin's equivalent circuit can be easily interchanged.

Exercise:

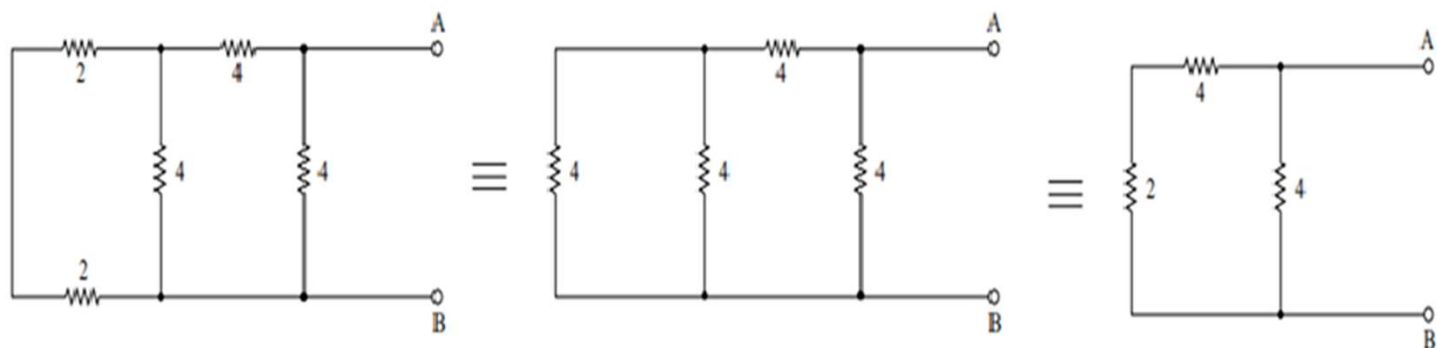
Applying Norton's theorem, let us calculate the current that would flow through the load resistance, $R_L = 5 \text{ ohm}$ as in Fig.. The first step is to remove the load resistance and then short circuit the terminals AB as shown in Fig., and calculate I_{SC} . This is done as follows.



By observation we see that when terminals A and B are shorted, terminals C and D also get shorted. Two $4 \text{ } \Omega$ resistances get connected in parallel across terminals EF, their equivalent resistance become $2 \text{ } \Omega$. Thus the total current, I supplied by the battery becomes



$$I = \frac{12}{2+2+2} = 2 \text{ A}$$



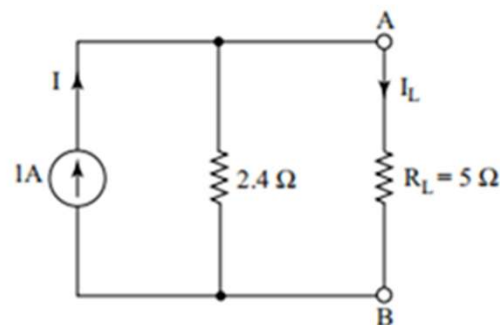
At point E, this $I = 2\text{ A}$ gets divided equally; 1 A going in branch EF and 1 A to branch EC (CD being shorted). At point C, 1 A current will flow through the short-circuited path provided between terminals A and B. Therefore, $I_{sc} = 1\text{ A}$.

Now, the resistance of the network viewed from the terminals AB when the battery is short circuited is

$$R_{AB} = \frac{6 \times 4}{10} = 2.4\ \Omega \quad (\text{see also Fig. 2.76})$$

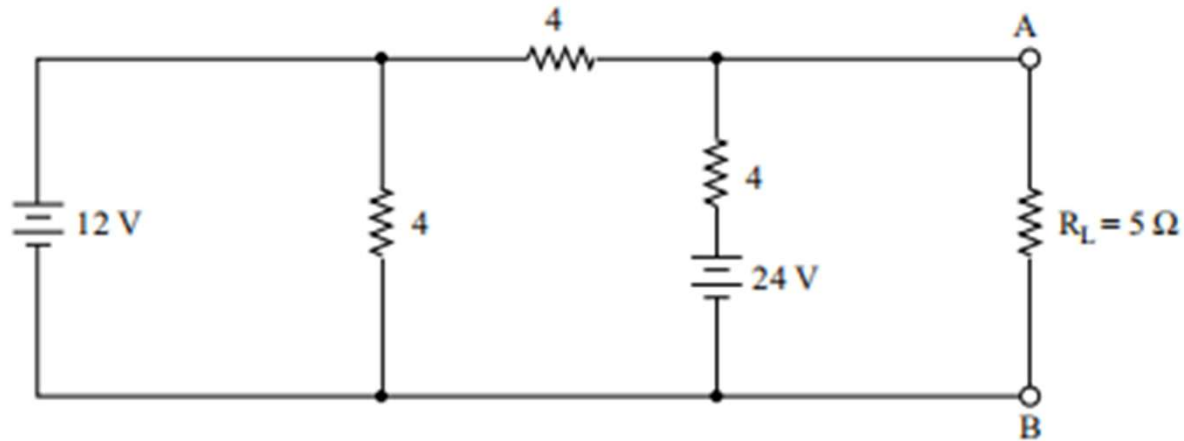
Norton's equivalent circuit is shown in Fig. 2.77. Now, using the current divider rule, the current through the load resistance is calculated as,

$$\begin{aligned} I_L &= I \frac{2.4}{2.4 + 5} \\ &= \frac{1.0 \times 2.4}{7.4} = 0.324\text{ A} \end{aligned}$$



Problem:

Using Norton's theorem calculate the current flowing through the load resistance connected across the terminals A and B as shown in Fig. Also apply Thevenin's theorem to calculate the same.



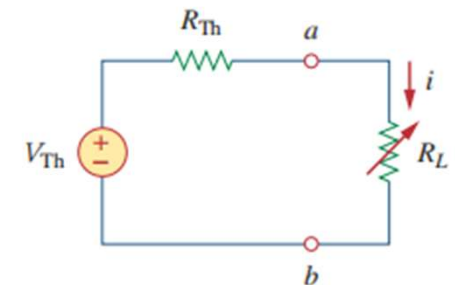
Maximum Power Transfer:

In many practical situations, a circuit is designed to provide power to a load. There are applications in areas such as communications where it is desirable to maximize the power delivered to a load.

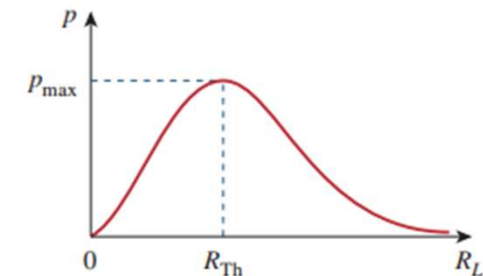
We now address the problem of delivering the maximum power to a load when given a system with known internal losses. It should be noted that this will result in significant internal losses greater than or equal to the power delivered to the load.

The Thevenin equivalent is useful in finding the maximum power a linear circuit can deliver to a load. We assume that we can adjust the load resistance. If the entire circuit is replaced by its Thevenin equivalent except for the load, as shown in Fig. (a), the power delivered to the load is

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad \dots\dots\dots(1)$$



For a given circuit, V_{Th} and R_{Th} are fixed. By varying the load resistance the power delivered to the load varies as sketched in Fig. (b). We notice from Fig. (b) that the power is small for small or large values of R_L but maximum for some value of R_L between 0 and ∞ . We now want to show that this maximum power occurs when R_L is equal to the R_{Th} . This is known as **Maximum power theorem**.



Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).

$$\begin{aligned}\frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0\end{aligned}$$

$$\boxed{R_L = R_{Th}} \dots\dots\dots(2)$$

The maximum power transferred is obtained by substituting R_L

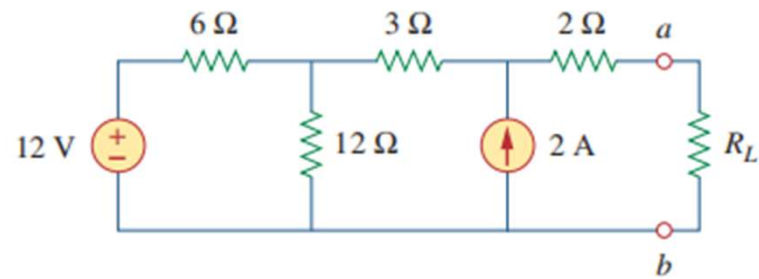
$$\boxed{p_{max} = \frac{V_{Th}^2}{4R_{Th}}} \dots\dots\dots(3)$$

Equation (3) applies only when $R_L = R_{Th}$.

When $R_L \neq R_{Th}$ we compute the power delivered to the load using Eq. (1).

Example:

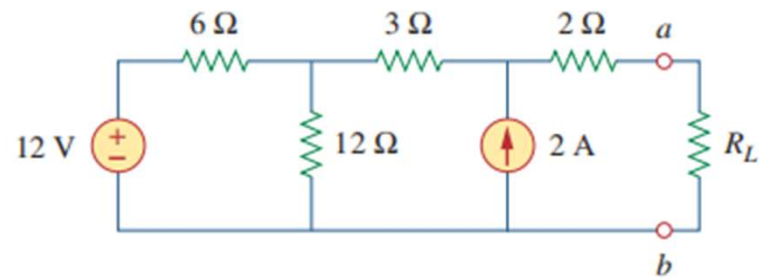
Find the value of R_L for maximum power transfer in the circuit of Fig. Find the maximum power.



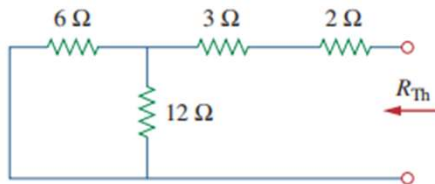
Solution:

Example:

Find the value of R_L for maximum power transfer in the circuit of Fig. Find the maximum power.

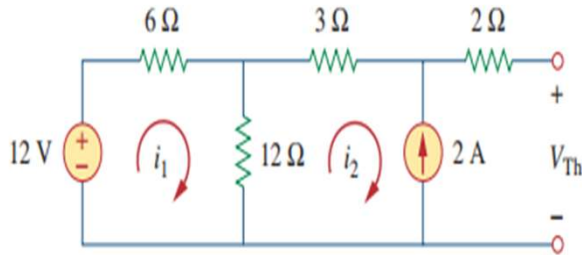
**Solution:**

Finding R_{Th}



$$R_{Th} = 2 + 3 + 6 \parallel 12 = 5 + \frac{6 \times 12}{18} = 9 \Omega$$

Finding V_{Th}



To get V_{Th} , we consider the circuit in Fig. (b). Applying mesh analysis gives

$$-12 + 18i_1 - 12i_2 = 0, \quad i_2 = -2 \text{ A}$$

Solving for i_1 , we get $i_1 = -2/3$. Applying KVL around the outer loop to get V_{Th} across terminals $a-b$, we obtain

$$-12 + 6i_1 + 3i_2 + 2(0) + V_{Th} = 0 \quad \Rightarrow \quad V_{Th} = 22 \text{ V}$$

For maximum power transfer,

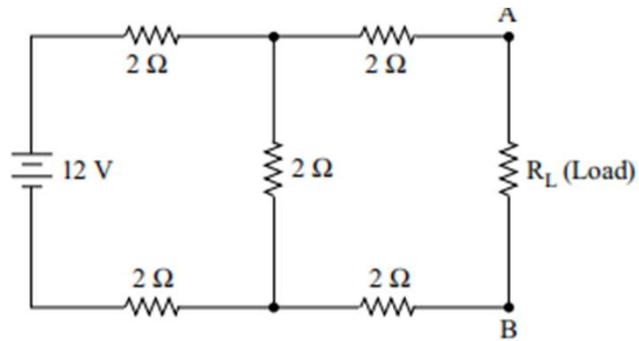
$$R_L = R_{Th} = 9 \Omega$$

and the maximum power is

$$p_{\max} = \frac{V_{Th}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44 \text{ W}$$

PROBLEM:

A 12 V battery is supplying power to a resistive load R_L through a network as shown in Fig. Calculate at what value of R_L power transferred to the load will be maximum and what would be the value of that maximum power.



References

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2. S. K. Bhattacharya," Basic Electrical and Electronics Engineering", Pearson Publication.
3. John Hiley, Keith Brown and Ian McKenzie Smith, "HUGHES ELECTRICAL AND ELECTRONIC TECHNOLOGY", tenth edition, Pearson Education Ltd. 2008.